Quantification: 872_wt_hp_z04
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	18
interior lacuna count	15
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 15)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2158.27	2077	1026.49	px^2
major_axis_length_px	lacuna shape	102.962	87.4647	60.9608	px
minor_axis_length_px	lacuna shape	32.7444	31.8422	9.5632	px
aspect_ratio	lacuna shape	3.5463	2.5698	2.9314	unitless
eccentricity	lacuna shape	0.9017	0.9212	0.083	unitless
solidity	lacuna shape	0.7333	0.7528	0.0933	unitless
roots_count	headline	13.3333	12	5.5891	count
ring_length_r30_px	headline	557.533	501	240.203	px
ring_length_r60_px	reported	1145.07	1137	588.644	px
owned_length_px	ownership-dependent	2209.15	2108.23	1622.01	px
edge_count	ownership-dependent	145.733	122	110.141	count
mean_edge_length_px	ownership-dependent	15.3594	15.116	1.8699	px
perimeter_px	normalised	313.163	273.321	146.578	px
ring_area_r30_px2	normalised	9345.27	8984	3499.87	px^2
ring_area_r60_px2	normalised	19908.9	20758	6654.28	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9951	1	0.0188	unitless
ring_density_r30	normalised	0.0605162	0.0597709	0.0203926	px^-1
ring_density_r60	normalised	0.0568577	0.051861	0.0232819	px^-1
roots_per_100px_perimeter	normalised	4.613	4.8447	1.9772	per 100 px
ring_attached_length_r30_px	network v2	523.2	466	243.132	px
ring_attached_length_r60_px	network v2	1050.4	1029	609.771	px
ring_length_w_r30_px	network v2	635.994	569.132	275.921	px
ring_length_w_r60_px	network v2	1304.31	1271.38	675.873	px
sholl_crossings_r10	network v2	12.5333	11	5.9506	count
sholl_crossings_r20	network v2	12	13	6.7823	count
sholl_crossings_r30	network v2	15.7333	15	8.0309	count
loop_count	loops, report only	21.4667	8	24.5847	count
loop_edge_share	loops, report only	0.4238	0.4706	0.2089	share
ring_width_mean_r30_px	width	7.0341	7.1239	0.3822	px

Per field
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measure	value	units
analysed_area_px2	864897	px^2
total_skeleton_length_px	35821	px
canalicular_length_density_per_px	0.0414165	px^-1
junction_count	1308	count
junction_density_per_px2	0.00151232	px^-2
lacuna_count	18	count

lacunae_per_px2	2.081e-05	px^-2
skeleton_component_count	122	count
mean_component_length_px	293.615	px
median_component_length_px	13.5	px
field_length_density_w_per_px	0.0470261	px^-1
field_density_without_flagged_per_px	0.0364895	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	70	px
unattached_component_count	5	count
canalicular_length_density_attached_per_px	0.0414165	px^-1
field_length_density_with_unattached_per_px	0.0414974	px^-1
excluded_area_px2	143103	px^2
analysed_field_area_px2	905473	px^2
loop_count	473	count
loop_edge_share	0.5928	share
edge_count	2741	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6665	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	36638	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3185	293.107	22.7028	12.9106	0.997	0.5399	-1.5651	143.896	232.375	False
2	2077	144.059	25.2241	5.7112	0.9846	0.6281	1.3769	151.947	448.976	False
3	4256	158.774	41.7686	3.8013	0.9648	0.6782	-1.4497	152.645	678.274	False
4	1429	52.067	37.8414	1.3759	0.6869	0.8279	-0.6051	154.502	593.396	False
5	1593	70.4006	31.8422	2.2109	0.8919	0.7622	1.3948	164.237	538.472	False
6	922	97.7946	15.9727	6.1226	0.9866	0.6272	1.3884	181.894	939.311	False
7	3025	100.943	44.409	2.273	0.898	0.65	-1.4641	211.119	126.158	False
8	2144	87.4647	34.0354	2.5698	0.9212	0.7528	-1.2557	226.871	777.66	False
9	2841	103.213	37.4578	2.7554	0.9318	0.8351	-1.3658	276.205	420.61	False
10	3125	68.6713	60.8024	1.1294	0.4648	0.909	0.622	332.681	995.111	True
11	3954	114.494	48.2385	2.3735	0.9069	0.7517	-1.4049	393.248	55.5005	True
12	3573	94.8523	52.0807	1.8213	0.8358	0.8284	-1.4851	395.903	280.417	False
13	1409	76.4593	28.6406	2.6696	0.9272	0.7683	-1.5183	392.216	831.581	False
14	2226	76.9066	39.7827	1.9332	0.8558	0.7987	-1.558	506.567	185.868	False
15	1503	75.7312	26.0285	2.9095	0.9391	0.8628	-1.3051	568.045	702.943	False
16	1123	61.8357	30.8107	2.007	0.867	0.6785	-1.3397	588.985	20.1576	True
17	628	38.8866	23.9678	1.6225	0.7875	0.7235	-1.4785	619.033	332.451	False
18	1563	73.771	29.411	2.5083	0.9171	0.717	-1.5252	651.639	225.151	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	19	668	1102	1659.8	122	13.6049	6.5894
2	8	344	557	549.333	36	15.2593	6.6102
3	11	462	611	1067.82	72	14.8308	7.3547
4	4	181	283	208.782	17	12.2813	7.1239
5	5	188	330	1395.23	82	17.015	7.012
6	12	302	512	347.764	20	17.3882	6.5629
7	23	840	1399	1919.47	115	16.6911	7.2004
8	18	756	1418	3064.84	207	14.806	7.2133
9	14	754	1743	2169.29	139	15.6064	7.206
10	13	358	847	1304.12	86	15.1641	7.5825
11	17	607	1362	1370.58	94	14.5807	7.3359
12	17	874	2140	4776.64	316	15.116	7.846
13	12	409	1137	2198.56	126	17.4489	6.4577
14	20	783	1870	2108.23	171	12.3288	7.3483
15	9	501	1061	2147.68	114	18.8393	6.8075
16	12	482	1240	2316.1	155	14.9426	6.8827
17	11	499	1252	6156.17	413	14.906	6.8692
18	17	802	1761	3367.6	236	14.2695	7.3103

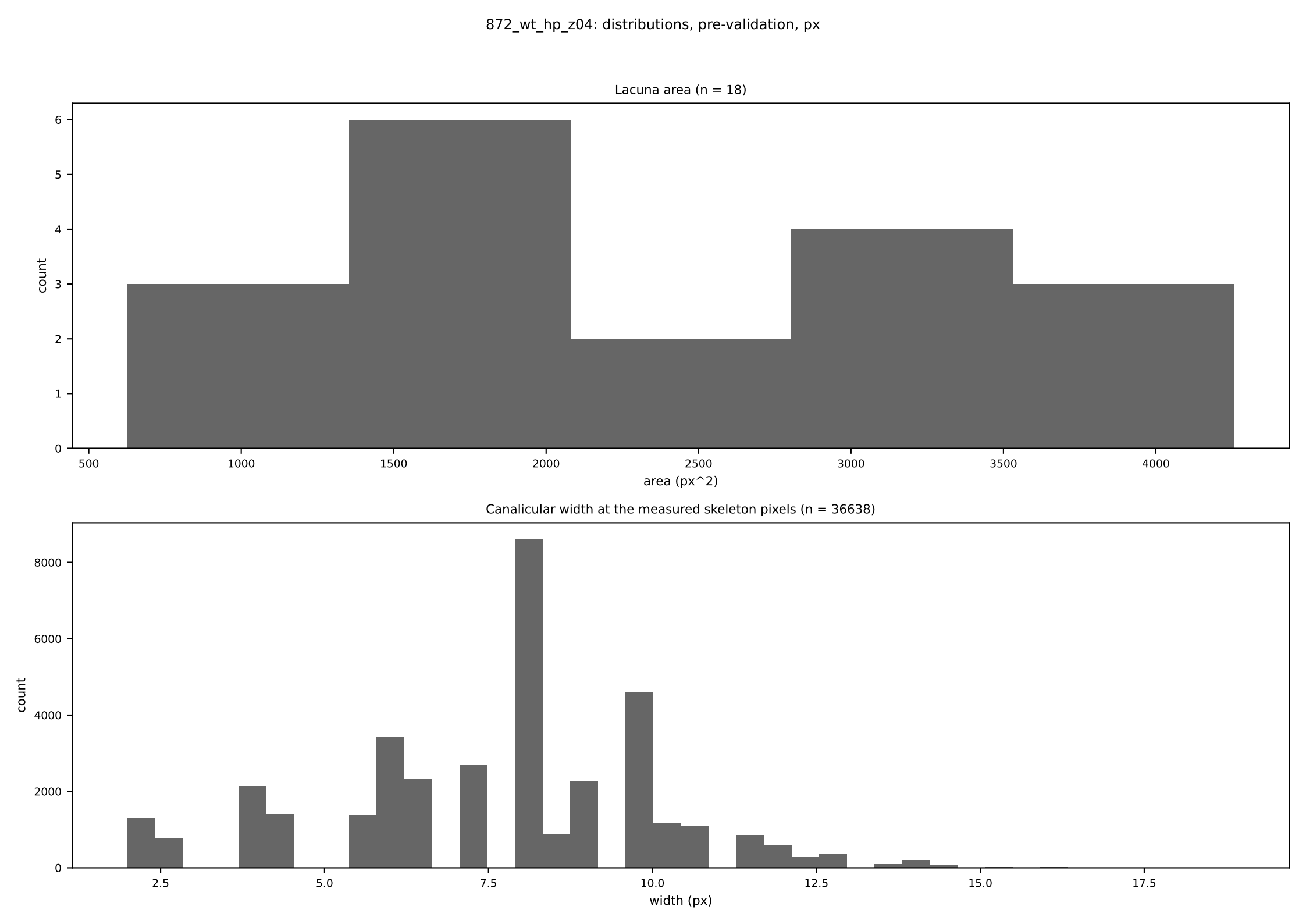
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	728.713	18655	35000	1	1	0.0358081	0.0314857	2.6073
2	364.67	8953	15872	1	1	0.0384229	0.0350932	2.1938
3	505.132	12229	21331	1	1	0.0377791	0.0286438	2.1776
4	197.238	3616	7483	1	1	0.0500553	0.0378191	2.028
5	234.723	4437	8224	1	1	0.042371	0.0401265	2.1302
6	237.723	8984	20267	1	0.9271	0.0336153	0.0252627	5.0479
7	425.345	10604	20758	1	1	0.0792154	0.0673957	5.4074
8	273.321	9813	19246	1	1	0.0770407	0.0736776	6.5857
9	288.978	10581	24146	1	1	0.0712598	0.0721859	4.8447
10	236.551	6707	15609	0.711	0.6585	0.0533771	0.0542636	5.4956
11	432.996	10444	22425	0.8508	0.7745	0.0581195	0.0607358	3.9261
12	315.191	11229	25289	1	1	0.0778342	0.0846218	5.3936
13	256.208	8359	21924	1	1	0.0489293	0.051861	4.6837
14	273.764	9314	21493	1	1	0.084067	0.0870051	7.3055
15	204.445	8382	22359	1	1	0.0597709	0.0474529	4.4022
16	216.752	5820	13951	0.7153	0.655	0.0828179	0.0888825	5.5363
17	148.61	6134	15204	1	1	0.0813499	0.0823467	7.4019
18	243.386	8889	20037	1	1	0.0902239	0.0878874	6.9848

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	617	1029	759.014	1243.99	20	13	11
2	259	533	388.037	639.76	6	4	10
3	455	611	536.274	710.571	11	8	10
4	181	246	206.711	320.723	3	3	4
5	162	238	216.095	374.25	5	2	5
6	270	332	339.007	575.345	7	6	9
7	809	1173	968.737	1605.09	22	15	16
8	679	1360	866.725	1617.25	17	16	25
9	732	1683	863.654	1997.43	18	19	22
10	308	675	399.635	964.708	7	12	12
11	564	1127	706.683	1562.71	14	14	14
12	854	2137	998.767	2464.55	19	25	31

13	342	940	458.605	1271.38	9	12	13
14	779	1853	884.441	2140.3	14	17	27
15	459	740	569.132	1182.64	11	7	15
16	475	1239	544.676	1418.12	12	12	14
17	466	1147	565.132	1416.15	10	14	20
18	784	1734	919.583	2005.28	16	19	18


872_wt_hp_z04: distributions, pre-validation, px
Lacuna area (n = 18)
count
area (px^2)
0
1
2
3
4
5
6
500
1000
1500
2000
2500
3000
3500
4000
Canalicular width at the measured skeleton pixels (n = 36638)
count
width (px)
0
2000
4000
6000
8000
2.5
5.0
7.5
10.0
12.5
15.0
17.5